



When Laplacian-Pyramid Decoding Equals Linearized Instant-NGP Interpolation

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Abstract

Under explicit conditions, the reconstruction obtained from a Laplacian pyramid with a separable tent is pointwise equivalent under the stated conditions, to that of an Instant-NGP-style multiresolution hash grid when per-level outputs are combined through a linear decoder and hashing is collision-free. In this linear regime, both synthesize signals as sums of shifted and dilated tensor-hat basis functions on a dyadic lattice. This highlights a structural bridge between classical multiresolution analysis and modern neural encodings.

1 Introduction

The Laplacian pyramid reconstructs images by iteratively expanding each level with a separable kernel. In the standard formulation, the EXPAND operator upsamples and filters the coarser level using the Burt–Adelson generating kernel. Decoding proceeds by expand-then-sum across levels [1]. At the parameter value $a = 0.5$, this operator becomes exactly a separable linear (triangular/tent) interpolation.

Instant-NGP, by contrast, computes at each level a d-linear interpolation of trainable per-vertex features on a grid. The per-level results are typically concatenated and processed by a small MLP [2]. Here, we replace the MLP by a linear sum across levels to isolate the purely linear synthesis.

Under dyadic scaling, aligned grids, and collision-free hashing (dense grid), the reconstruction produced by both methods is equivalent pointwise. This observation concerns the synthesis (decoding) operator only: when the EXPAND operator uses $a = 0.5$, the effective interpolation weights coincide with the separable d-linear tent basis used in Instant-NGP.

2 Assumptions

- **Domain and boundaries.**

Periodic domain $\Omega = [0, 1)^d$. All sums are taken with wrap. With zero padding, the equalities hold pointwise only on the interior.

- **Dyadic lattice.**

In Instant-NGP [2], the grid resolutions across levels follow a geometric progression

$$N_\ell = \lfloor N_{\min} b^\ell \rfloor,$$

with a typical growth factor $b \in [1.26, 2]$. This choice ensures overlapping coverage and gradual refinement rather than exact dyadic doubling. In this work, we specialize to the dyadic case $b = 2$ and assume phase-aligned grids across levels. Under these conditions, each level’s lattice coincides with that of a classical multiresolution analysis, which allows a direct pointwise equivalence with the Laplacian pyramid synthesis operator.

- **Interpolation kernel.**

Separable tent basis:

$$\Phi(u) = \prod_{i=1}^d \max(1 - |u_i|, 0). \quad (2.1)$$

Instant-NGP uses d -linear interpolation at each level, which coincides with Φ in Eq. (2.1) when written in basis form [2]. For the Laplacian pyramid, setting the Burt–Adelson parameter to $a = 0.5$ makes the EXPAND operator equivalent to tent interpolation (triangular weighting) [1].

Burt and Adelson’s generating kernel in one dimension is parameterized by a coefficient a as

$$\hat{w}(0) = a, \quad \hat{w}(\pm 1) = \frac{1}{4}, \quad \hat{w}(\pm 2) = \frac{1}{4} - \frac{a}{2},$$

under the equal-contribution constraint described in [1]. At $a = \frac{1}{2}$ this yields weights

$$\hat{w} = [0, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0],$$

whose corresponding EXPAND operator (upsample + filter) implements linear B-spline interpolation. In the continuous basis view, this reconstruction kernel is exactly the tent (degree-1 B-spline) function

$$h(u) = \max(1 - |u|, 0),$$

applied separably across dimensions:

$$\Phi(u) = \prod_{i=1}^d h(u_i).$$

Hence, when the Laplacian pyramid uses $a = \frac{1}{2}$, its EXPAND operator is equivalent to the separable tent interpolant also used in the linear Instant-NGP grid.

- **Aligned grids.**

All levels lie on the same dyadic lattice. There are no per-level shifts.

- **Hashing (collision-free).**

At level ℓ , injectivity of the corner-index map holds. Under the periodic wrap, we require $N_\ell^d \leq T$ (unique vertices = N_ℓ^d). With open boundaries use $(N_\ell + 1)^d \leq T$. In this regime, each level behaves as a dense grid.

- **Linear reduction across levels.** We replace the MLP by a linear sum across levels so the overall decoder is linear.

3 Equivalence of the synthesis operators

Notation and coordinates. We work on the unit cube $\Omega = [0, 1]^d$ with dyadic levels $N_\ell = 2^\ell$. Grid nodes are indexed by $k \in \mathbb{Z}^d$, and continuous coordinates $x \in \Omega$ are evaluated through the separable tent basis Φ of Eq. (2.1).

Tent basis expansion. For nodal coefficients $a_\ell[k]$ on the level- ℓ grid, define the continuous expansion (interpolant)

$$\text{Expand}^{(\ell)}(a_\ell)(x) = \sum_k a_\ell[k] \Phi(2^\ell x - k), \quad (3.1)$$

which corresponds to bi/tri/d-linear interpolation expressed in basis form (cf. Eq. (2.1)).

Here, the coefficients $a_\ell[k]$ represent the discrete values stored at the grid nodes of level ℓ . They correspond to the pixel samples $g_\ell(i, j)$ in the classical Laplacian pyramid and to the per-corner feature values in the Instant-NGP hash grid.

Hash-grid reconstruction. At level ℓ , Instant-NGP computes the d-linear interpolant (Fig. 3 in [2])

$$H_\ell(x) = \sum_k c_\ell[k] \Phi(2^\ell x - k). \quad (3.2)$$

In the *linear* variant considered here, the full reconstruction is the sum across levels:

$$\hat{f}_{\text{hash}}(x) = \sum_{\ell=0}^L H_\ell(x) = \sum_{\ell=0}^L \sum_k c_\ell[k] \Phi(2^\ell x - k). \quad (3.3)$$

which corresponds to a linear combination across levels.

In the practical Instant-NGP implementation each vertex stores an F -dimensional feature vector $f_\ell[k]$. A linear per-level readout $w_\ell \in \mathbb{R}^F$ produces effective scalar coefficients $c_\ell[k] = w_\ell^\top f_\ell[k]$. Under this linearization the synthesis in Eqs. (3.2) and (3.3) is identical to that obtained with scalar nodal values $c_\ell[k]$.

Laplacian-pyramid reconstruction. Let $L_\ell[k]$ denote Laplacian bands for $\ell = 0, \dots, L-1$ and $G_L[k]$ the coarsest Gaussian residual. Classical reconstruction proceeds by expand-then-sum:

$$\hat{f}_{\text{LP}}(x) = \sum_{\ell=0}^{L-1} \text{Expand}^{(\ell)}(L_\ell)(x) + \text{Expand}^{(L)}(G_L)(x), \quad (3.4)$$

i.e.,

$$\hat{f}_{\text{LP}}(x) = \sum_{\ell=0}^{L-1} \sum_k L_\ell[k] \Phi(2^\ell x - k) + \sum_k G_L[k] \Phi(2^L x - k), \quad (3.5)$$

which mirrors the expand-then-sum decoding identity in [1].

Both formulations can be viewed as expansions in a separable tent basis on a dyadic lattice. The Laplacian pyramid's coefficients $\{L_\ell[k], G_L[k]\}$ and the hash grid's per-level features $\{c_\ell[k]\}$ therefore play the same structural role as nodal coefficients of the same multiscale basis. The difference lies only in nomenclature: residual bands versus trainable features.

Observation. Under the assumptions above, the linear-decoder hash grid and the Laplacian pyramid with tent EXPAND produce identical reconstructions for the coefficient identification

$$c_\ell[k] = \begin{cases} L_\ell[k], & \ell = 0, \dots, L-1, \\ G_L[k], & \ell = L. \end{cases} \quad (3.6)$$

Proof. Substituting Eq. (3.6) into Eq. (3.3) and comparing termwise with Eq. (3.5) gives

$$\begin{aligned} \hat{f}_{\text{hash}}(x) &= \sum_{\ell=0}^L \sum_k c_\ell[k] \Phi(2^\ell x - k) \\ &= \sum_{\ell=0}^{L-1} \sum_k L_\ell[k] \Phi(2^\ell x - k) + \sum_k G_L[k] \Phi(2^L x - k) \\ &= \hat{f}_{\text{LP}}(x) \end{aligned} \quad (3.7)$$

□

Optimization equivalence. Because the two reconstructions coincide pointwise under the mapping Eq. (3.6), any differentiable loss $\mathcal{L}(\hat{f})$ induces identical gradients on the per-level coefficients:

$$\frac{\partial \mathcal{L}}{\partial c_\ell[k]} = \frac{\partial \mathcal{L}}{\partial L_\ell[k]}.$$

Consequently, optimizing each level of the Laplacian pyramid proceeds identically to optimizing the corresponding hash-grid parameters in the linear regime, which matches our empirical findings in Figure 1.

Table 1: Structural correspondence between the Laplacian pyramid and the linear hash-grid synthesis (under the linear and collision-free assumptions).

Concept	Laplacian Pyramid	Linear Hash Grid
Coefficient symbols	$L_\ell[k], G_L[k]$	$c_\ell[k]$
Interpolation kernel	EXPAND($a=0.5$)	d-linear Φ
Grid resolution	$N_\ell = 2^\ell$	$N_\ell = 2^\ell$
Combination rule	Expand-then-sum	Level-wise sum
Domain	Periodic image lattice	Periodic voxel lattice
Nonlinear stage	None	Removed MLP

This equivalence is purely structural: it holds pointwise and linearly under the stated assumptions. Deviations such as non-dyadic scaling, hash collisions, non-tent interpolation, or nonlinear decoders introduce departures from exact equality while preserving the same multiscale tent structure.

3.1 When the equivalence breaks

Exact equality holds only under the stated linear, dyadic, and collision-free assumptions. Deviations appear when:

1. $a \neq 0.5$: the Burt–Adelson kernel is no longer a tent, so the reconstruction kernel differs from the d-linear basis.
2. Hash collisions: non-injective vertex mapping causes gradient-averaged coefficients, producing an effective low-pass modulation [2].
3. Nonlinear decoding: an MLP introduces cross-level coupling, turning the linear sum into a nonlinear composition.
4. Non-dyadic or shifted grids: mis-aligned phase between levels invalidates the point-wise identity.
5. Boundary effects: for open boundaries the equality holds only on the interior unless padded.

These deviations are smooth perturbations of the same multiscale tent expansion.

4 Experimental verification

To validate the theoretical equivalence, we optimized a texture representation under both parameterizations: a Laplacian pyramid with $a = 0.5$ and a linearized Instant-NGP hash grid with collision-free dense grids. Both were trained to reproduce the same target image under an ℓ_2 loss on pixel intensities.

Implementations. For the Laplacian pyramid, we use the publicly available codebase of Weier et al. [3], configured as a flat mip-aware image pyramid with tent EXPAND ($a=0.5$). For the hash grid, we use the Mitsuba implementation of the multiresolution hash-grid encoding with the MLP removed and a linear level-wise sum as in Eq. (3.3). All experiments are run in Mitsuba using its CUDA automatic-differentiation backend.

Protocol. Each level’s coefficients are optimized independently with identical Adam hyperparameters and learning-rate schedules. For the hash grid, we use three features per level ($F=3$) corresponding to the RGB channels of the Laplacian pyramid, and a linear readout that directly maps them to color values, yielding effective nodal coefficients $c_\ell[k] \in \mathbb{R}^3$. For the Laplacian pyramid, we optimize the per-channel coefficients $\{L_\ell[k], G_L[k]\}$ directly. Both parameterizations are initialized to zero and use identical sampling budgets per iteration.

Results. We optimized a 4K texture of *The Great Wave off Kanagawa* for 128 iterations under an ℓ_2 reconstruction loss. The per-iteration loss curves of the Laplacian pyramid and the linear hash grid are indistinguishable within numerical precision, fully overlapping across the entire optimization. Both follow the same non-monotonic descent pattern and reach identical minima, confirming that their gradient flows and convergence dynamics are equivalent in practice. The final reconstructed textures are visually and numerically identical to machine precision, as predicted by the analytical gradient identity.

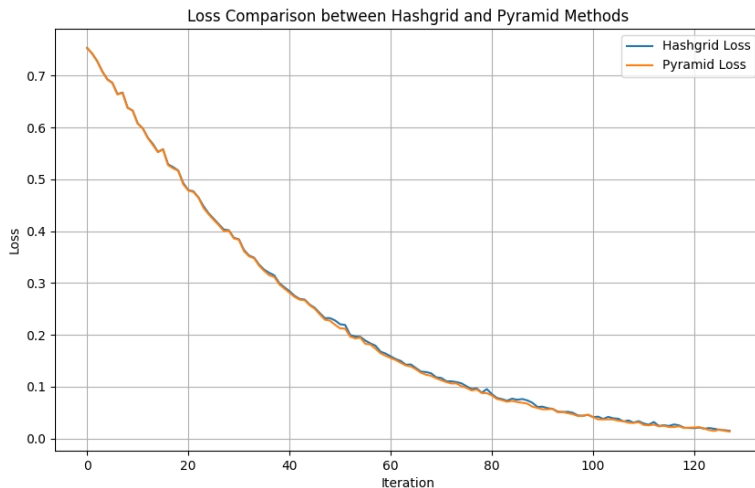


Figure 1: Loss evolution during the 4K texture optimization of *The Great Wave off Kanagawa* over 128 iterations. The Laplacian pyramid (orange) and the linear hash grid (blue) curves overlap exactly, confirming identical optimization dynamics and loss landscapes.

References

- [1] P. Burt and E. Adelson. The laplacian pyramid as a compact image code. *IEEE Transactions on Communications*, 31(4):532–540, 1983.
- [2] Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. Instant neural graphics primitives with a multiresolution hash encoding. *ACM Trans. Graph.*, 41(4):102:1–102:15, July 2022.
- [3] Philippe Weier, Jérémy Riviere, Ruslan Guseinov, Stephan Garbin, Philipp Slusallek, Bernd Bickel, Thabo Beeler, and Delio Vicini. Practical inverse rendering of textured and translucent appearance. *Transactions on Graphics (Proceedings of SIGGRAPH)*, 44(4), August 2025.